26900525 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCSTA300 - APPLIED REGRESSION ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **10** questions

Each question carries **3** marks

1. State any three common violations of regression assumptions. [U] / [CO1]
2. What is the role of residual plots in regression diagnostics? [An] / [CO1]
3. Define the coefficient of determination in regression analysis. [U] / [CO1]
4. Write the least-squares estimator of the intercept.coefficient in a regression model. [U] / [CO1]
5. State the role of the intercept in a multiple regression model. [U] / [CO2]
6. What is meant by efficiency of an estimator? [U] / [CO2]
7. State any two remedies for serial correlation. [U] / [CO2]
8. State any two limitations of the ordinary least squares (OLS) method in the presence of outliers. [An] / [CO3]

9. Define an orthogonal polynomial. [U] / [CO3]

10. Why are starting values important in nonlinear estimation? [U] / [CO4]

11. What is meant by logistic-regression diagnosis? [A] / [CO4]

12. Distinguish between fixed effects and random effects in a regression model. [U] / [CO5]

[3x10 = 30]

Part B

Short essay type questions

Answer any **4** questions

Each question carries **7** marks

13. Discuss the major applications of regression analysis. [An] / [CO1]

14. Explain the simple linear regression model and its components. [U] / [CO1]

15. Explain the need for model selection in multiple regression. [An] / [CO2]

16. Explain the t-test used for regression coefficients in multiple linear regression. [A] / [CO2]

17. Explain how GLS transforms the original regression model. [U] / [CO3]

18. Discuss the use of correlation matrices for detecting multicollinearity. [U] / [CO5]

[7x4 = 28]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

19. Compare parametric and nonparametric regression approaches in detail. [U] / [CO3]

20. Discuss different methods of transforming nonlinear relationships into linear forms. [An] / [CO4]

[12x1 = 12]